

Jo Choi

AI/ML Engineer

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EDUCATION

New York University — M.S., Computer Engineering	Fall 2024 – May 2026 (Expected)
Illinois Institute of Technology — B.S., Computer Engineering; Minor, AI	Fall 2022 – Spring 2024
Inha University — B.S., Electronic Engineering (Dual Degree)	Spring 2016 – Spring 2024

SKILLS

Languages: Python, C++, PostgreSQL, MySQL, SQLite

AI/ML Frameworks: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, Keras, MLflow

Data & Visualization: pandas, NumPy, Matplotlib, Power BI

Infrastructure & Deployment: Docker, Kubernetes, Amazon AWS (SageMaker), FastAPI, REST APIs, Linux, Git

WORK EXPERIENCE

Data Scientist Intern — C&B Trading, Seoul, Korea Jun 2024 – Aug 2024

- Collaborated with the sales and operations teams to develop a time-series forecasting model in Python (pandas, scikit-learn) for semiconductor component demand, achieving 90% accuracy and replacing manual spreadsheet estimates
- Automated weekly data consolidation from order records, supplier price sheets, and shipping logs using Python (pandas), reducing analyst prep time by 60%
- Designed interactive Power BI dashboards tracking supplier lead times, inventory turnover, and shipping delays across 3 regions, enabling management to renegotiate terms with 2 underperforming suppliers
- Analyzed pricing and margin data across 4 product categories, identifying optimal markup ranges that informed quarterly pricing strategy for cross-border transactions

RESEARCH EXPERIENCE & PROJECTS

Multimodal Gaze Heatmap Prediction from Egocentric Video

- Enable real-time gaze prediction on CPU for AR/VR and assistive-tech applications by building a compact multimodal model that fuses vision and audio signals.
- Built a 3.4M-parameter FiLM-based fusion model in PyTorch (~13 MB, ~2 GFLOPs) combining a ResNet-18 vision encoder with an audio CNN; trained on 110K+ samples from EGTEA Gaze+.
- Achieved 7.94 px mean gaze error on a 56×56 heatmap grid at ~40 ms CPU latency, enabling real-time inference without GPU acceleration.

Efficient Vision-Language Model (VLM) Design with Sparse Techniques

Advised by Prof. Sai Qian Zhang, NYU

- Cut the inference cost of vision-language models by selecting only the most salient visual tokens while preserving accuracy on visual reasoning benchmarks.
- Architected a saliency-based VLM pairing a Vision Transformer with an LLM via a projection layer for adaptive token selection; reproduced SparseVLM on LLaVA-1.5-7B and ran ablations against weight-pruning methods.
- Reduced visual tokens by 30% with minimal accuracy loss on VQAv2, outperforming three weight-pruning baselines.

Gender Bias & Rating Prediction in Higher Education Evaluations

- Quantify gender bias in student-written professor evaluations and predict ratings from review text to inform fairer faculty-review practices.
- Analyzed a 90K-row evaluation dataset in Python; built Ridge regression and logistic regression pipelines with cross-validated hyperparameters for rating prediction and gender classification.
- Achieved $R^2 = 0.53$ for rating prediction and ROC-AUC = 0.77 for gender classification, surfacing statistically significant bias terms in evaluation language.

Multimodal Mouse Behavior Recognition using Deep Learning

Advised by Prof. Yan Yan, Illinois Institute of Technology

- Automate mouse-behavior classification from synchronized video and audio to replace manual annotation and scale up neuroscience and pharmacology experiments.
- Developed an RNN-based multimodal classifier in PyTorch fusing visual features from video frames with audio from synchronized microphones; trained on a labeled behavior dataset.
- Achieved 90% accuracy across six behaviors, replacing manual annotation for the lab's downstream experiments.

AWARDS & ACTIVITIES

Teaching Assistant, New York University — ECE-UY 3114 Fundamentals of Electronics I	Sep–Dec 2025
President & Founder, Korean Student Association at Illinois Institute of Technology	Jun 2023–May 2024
Active Member, Triangle Fraternity (Armour Chapter)	Aug 2022–May 2024
Dean's List at Illinois Institute of Technology	Spring 2023
Honor Student & Merit Scholarship at Inha University	Fall 2019
ROK Military — Honorably discharged as Sergeant / Squad Leader (21 months)	Aug 2017–Apr 2019